



EE 396 DESIGN LAB

Supervisor: Dr. Shabari Nath

Project: BUCK CONVERTER

Group: 70

STUDENTS:

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Department: EEE

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Components Used:

Sr. No.	MPN	Description	Qty issued	Soldered on PCB	Not Used
1	24.247.1 / MULTICOMP	Banana Socket Panel Red	2	2	0
2	ECWF2W225P8 / PANASONIC	Film Capacitor 2.2 μ F 450V	2	2	0
3	IRF540NPBF / INFINEON	Power MOSFET 100V 33A	1	1	0
4	352215RFT / CGS-TE CONNECTIVIT	SMD Resistor 15 Ω 3W	1	1	0
5	355010RFT / CGS-TE CONNECTIVIT	SMD Resistor 10 Ω 5W	2	2	0
6	FKP2D010001D00JS / WIMA	Film Capacitor 1000pF	2	2	0
7	SHBC24N-2R3A0104V / KEMET VS-20TQ045-M3	Power Inductor 104 μ H	1	1	0
8	Vishay	Schottky Diode 45V 20A	1	1	0
9	EEUFS2A330B / PANASONIC	Electrolytic Capacitor 33 μ F 100V	1	1	0
10	MCWR06X1203FTL / MULTICOMP	Resistor 120k 0603	1	0	1
11	ERJ-3GEYJ125V / PANASONIC	Resistor 1.2M 0603	2	0	2
12	22-28-4360 / MOLEX	Pin Header 36 Way	2	Very few pins used	2
13	FK243MI2470 / FISCHER	Heat Sink TO-220	2	2	0
14	HCPL-3120-300E / BROADCOM	Optocoupler Gate Driver	1	1	0
15	MCWR06X101JTL / MULTICOMP	Resistor 100 Ω 0603	1	1	0
16	06035C104JAT2A / KYOCERA	MLCC 0.1 μ F 0603	6	6	0
17	TBA2-1222 / TRACO POWER	DC/DC Converter 12V	1	1	0
18	KPT-1608LVSECK-J3-PRV / KINGBRIGHT	LED Red 0603	2	0	2
19	1715721 / PHOENIX CONTACT	Terminal Block 2 Way	1	1	0

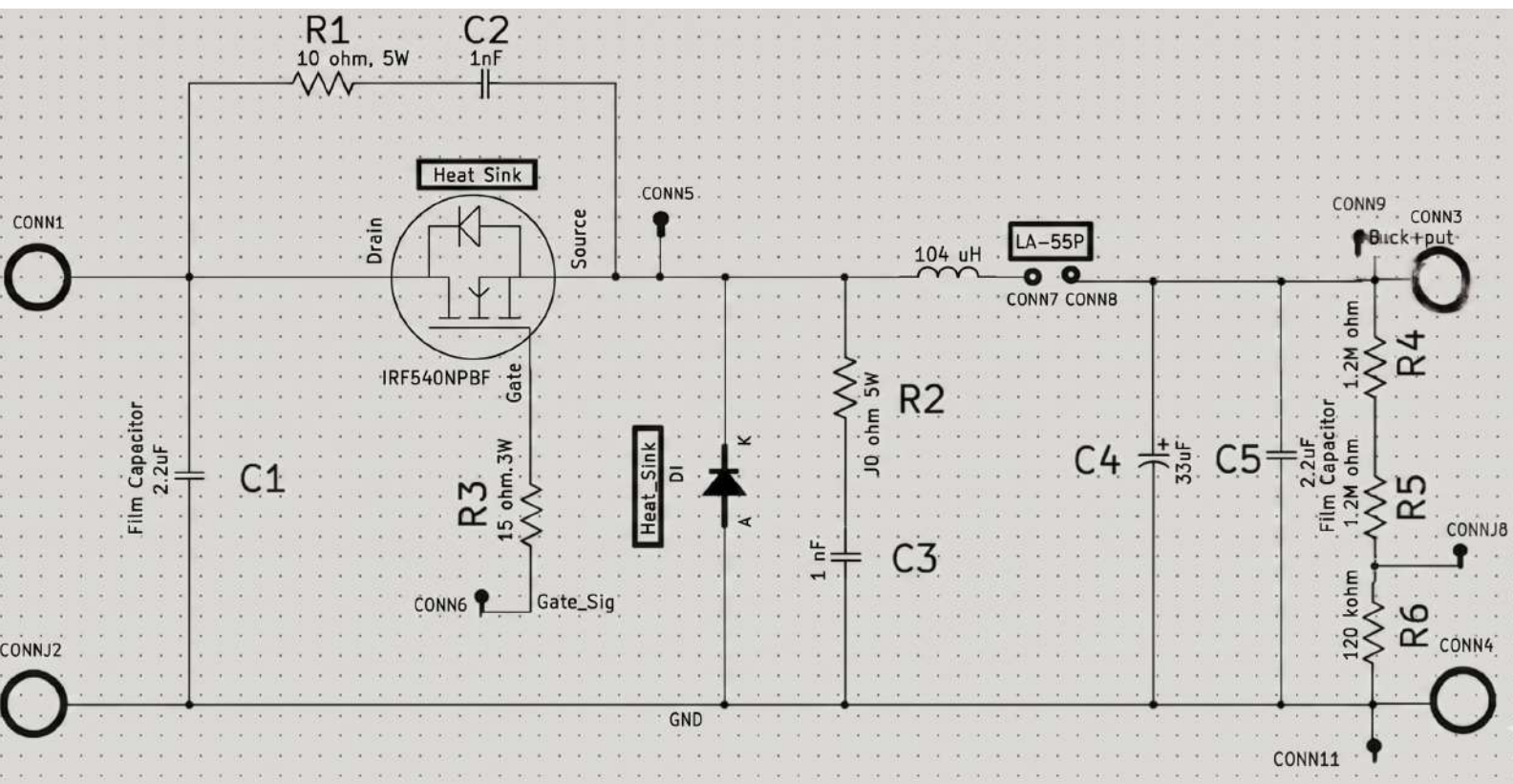
20	MCPWR05FTW 0750 / MULTICOMP	Resistor 75Ω 0805	1	0	1
21	ERJ- PA3F4752V /	Resistor 47.5k 0603	2	0	2
22	031-5329-52RFX / AMPHENOL	BNC Jack 50Ω	2	0	2
23	MCWR08X2002 FTL / MULTICOMP	Resistor 20k 0805	2	2	0
24	MCWR12X1003 FTL / MULTICOMP	Resistor 100k 1206	4	4	0
25	LF347DT / STMicroelectroni	Op Amp LF347	2	0	2
26	MMSZ4710T1G / onsemi	Zener Diode 25V	5	0	5
27	LA55-P / LEM	Current Transducer 50A	1	1	0
28	RT1206BRD075 KL / YAGEO	Resistor 5k 1206 0.1%	6	1	5

PCB Label Mapping with Corresponding R and C value:

Resistor	Value	Capacitor	Value
R1	10 ohm,5W	C1	2.2 uF
R2	10 ohm,5W	C2	1 nF
R3	15 ohm, 3W	C3	1 nF
R4	1.2Mohm (Not used)	C4	33 uF
R5	1.2M ohm (Not used)	C5	2.2 uF
R6	120k ohm (Not used)	C6	0.1 uF
R7	100 ohm	C7	0.1 uF
R8	5k ohm	C8	0.1 uF
R9	5k ohm (Not used)	C9	0.1 uF
R10	5k ohm (Not used)	C10	0.1 uF
R11	5k ohm (Not used)	C11	0.1 uF
R12	5k ohm (Not used)		
R13	5k ohm (Not used)		
R14	47.5k ohm (Not used)		
R15	47.5k ohm (Not used)		
R16	75 ohm (Not used)		
R17	100k ohm		

R18	100k ohm		
R19	100k ohm		
R20	100k ohm		
R21	20k ohm		
R22	20k ohm		

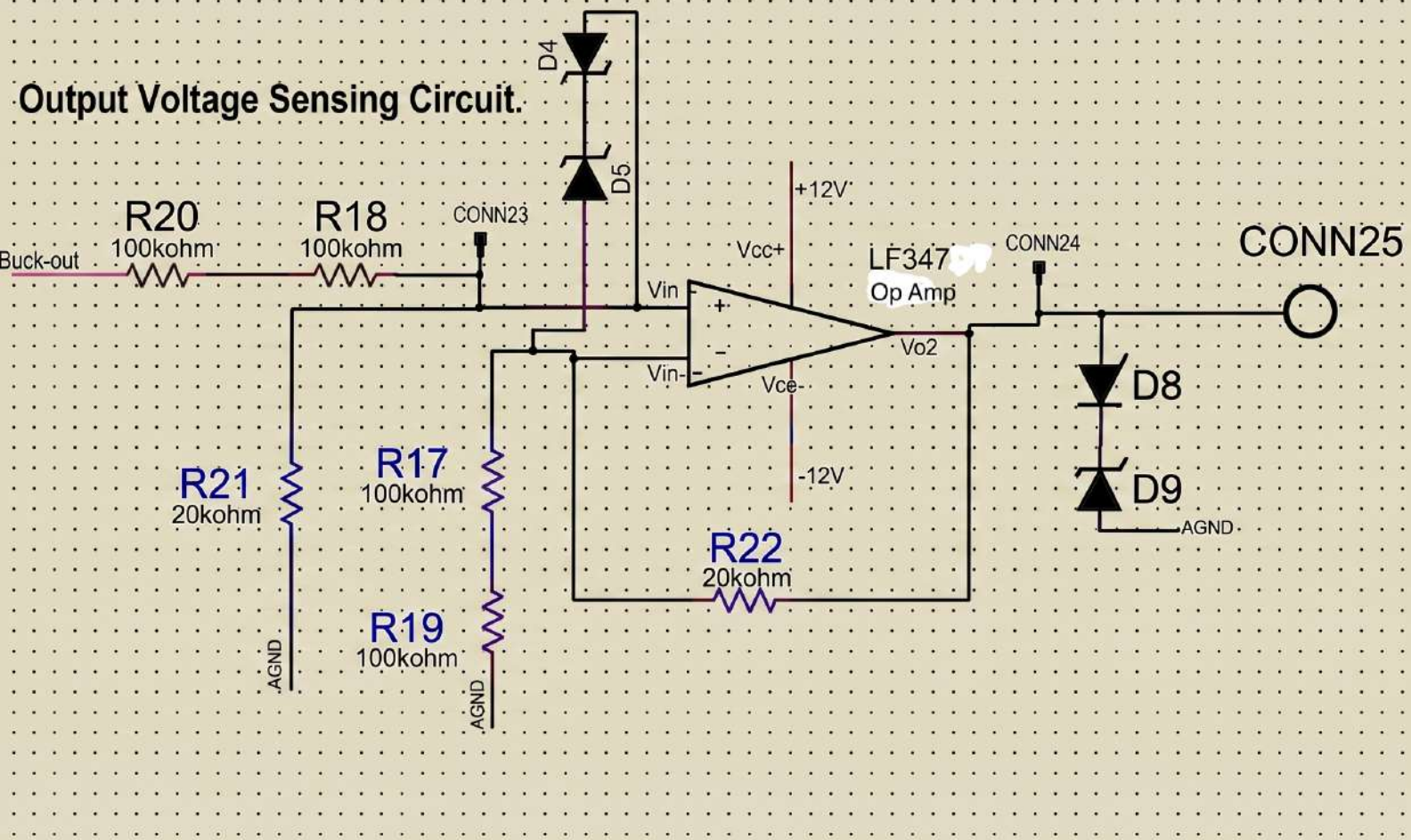
Buck Main Circuit:



Working of Buck main circuit:

The main buck converter circuit steps down the input DC voltage to a lower regulated output using high-frequency switching. When the MOSFET (IRF540N) is turned ON by the gate signal from the driver circuit, the input voltage is directly applied across the inductor ($104\mu\text{H}$), causing current through the inductor to increase linearly and energy to be stored in its magnetic field. During this interval, the diode is reverse-biased and does not conduct. When the MOSFET turns OFF, the inductor resists the sudden drop in current and releases its stored energy, forward-biasing the diode and supplying current to the load. This continuous transfer of energy maintains current flow even when the switch is OFF. The output capacitors (C4 and C5) smooth out the voltage and reduce ripple, providing a stable DC output. The input capacitor (C1) filters input voltage fluctuations. The RC network (R1–C2 and R2–C3) helps in damping switching noise and reducing voltage spikes, protecting the MOSFET and improving stability. The inductor ensures continuous current flow, which is the key reason for reduced ripple and improved efficiency in buck converters. The resistor network (R4, R5, R6) forms a voltage divider used for sensing or feedback purposes. By controlling the duty cycle of the MOSFET switching through the gate signal, the average output voltage is regulated to the desired lower value.

Voltage Sensing Circuit:



Working of Voltage Sensing Circuit:

The buck converter output voltage (maximum 30V) is first reduced using a resistor divider network. The effective division is designed such that the voltage is scaled down by a factor of **0.1**, i.e.

$$V_{div} = V_{out} \times (20k / (20k + 200k)) \approx V_{out} \times 0.091$$

This reduced voltage is applied to the **non-inverting input of the op-amp (LF347)**. The op-amp is configured in a non-inverting configuration with feedback resistors such that its gain is:

$$\text{Gain} = 1 + 20k/200k = 1.1$$

Therefore, the overall transfer from output voltage to sensed voltage becomes:

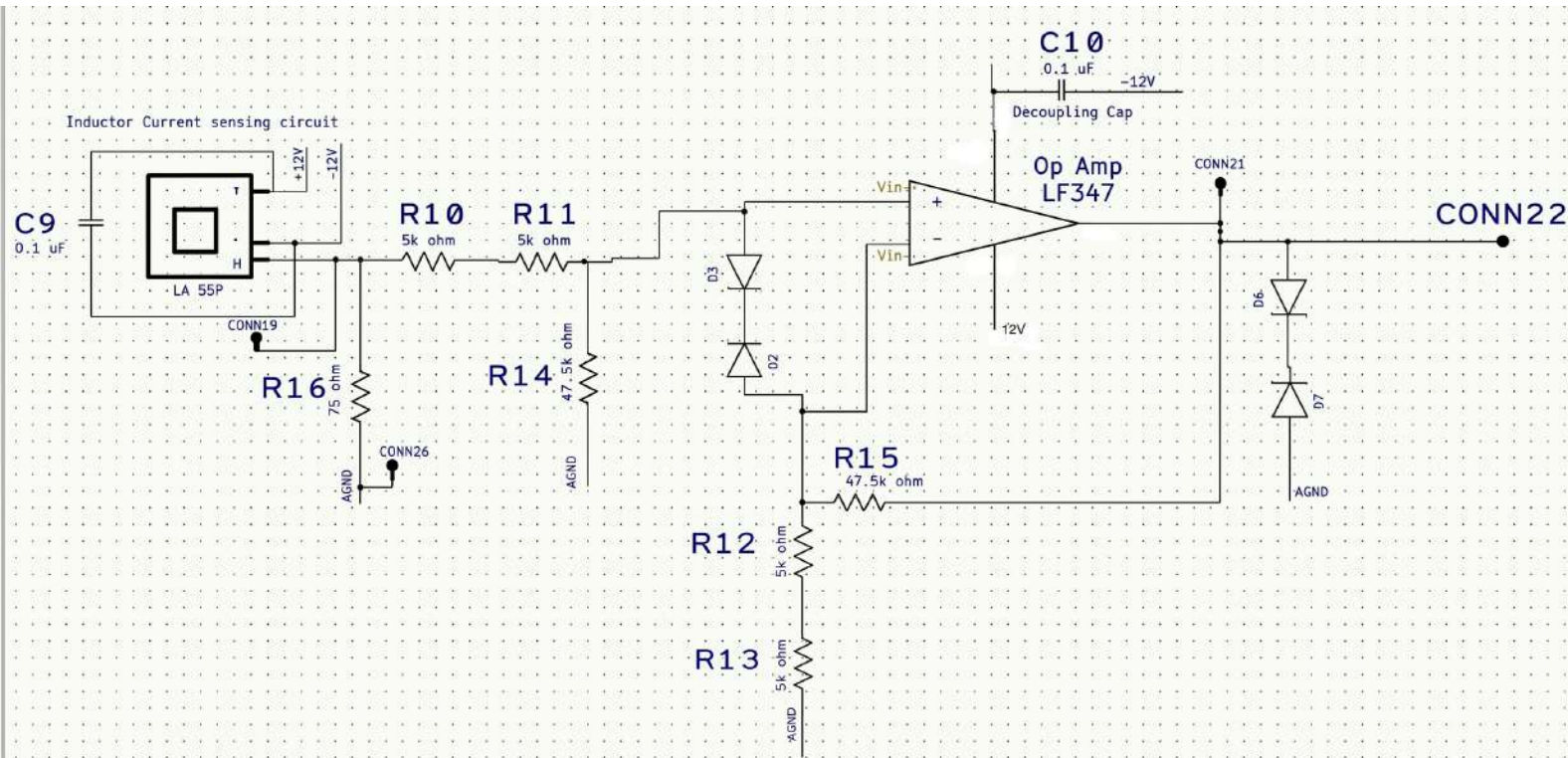
$$V_{sense} = V_{out} \times 0.091 \times 1.1 \approx V_{out} \times 0.1$$

For the maximum output voltage of **30V**, the final sensed voltage is:

$$V_{sense} = 30 \times 0.1 = 3V$$

Hence, the circuit ensures that the sensed output voltage always remains within the **0–3.3V range**.

Inductor Current Sensing Circuit:



Working of Inductor Current Sensing circuit:

The inductor current is measured using a Hall-effect current transducer (LA-55P), which provides an isolated current measurement. The sensor converts the inductor current into a proportional secondary current given by:

$$I_s = I_L / 1000$$

This secondary current is passed through a burden resistor **R16 (75 Ω)** to convert it into a voltage:

$$V_s = I_s \times 75 = I_L \times (1/1000) \times 75$$

This voltage is then processed through a resistor network and op-amp stage for scaling and conditioning. The attenuation due to the divider formed by **R14** and series combination of **R10** and **R11** is: **47.5 / (47.5+10)**

The op-amp is configured as a non-inverting amplifier with gain:

$$\text{Gain} = 1 + 47.5/10 = 5.75$$

Combining all stages, the overall transfer relation becomes:

$$V_{out} = I_L \times (1/1000) \times 75 \times (47.5/57.5) \times 5.75$$

$$V_{out} \approx I_L \times 0.356$$

For maximum Inductor current of **8A**, the output voltage is:

$$V_{out} = 8 \times 0.356 = 2.85 \text{ V}$$

Thus, the circuit is designed such that the maximum inductor current of **8 A** is mapped to approximately **2.85 V**, which lies safely within the **0–3.3V range**, making it suitable for ADC measurement.

Calculations for Testing:

Input Voltage , $V_{in} = 30 \text{ V}$

Duty ratio , $D = 0.75$

Output Voltage , $V_{out} = V_{in} * D = 22.5 \text{ V}$

Switching frequency , $f = 100 \text{ KHz}$

Inductor , $L = 104 \mu\text{H}$

Ripple inductor current = $((V_{in} - V_o) * D) / (L * f) = 540 \text{ mA}$

Output Boundary current for CCM, $I_b = \text{Ripple current} / 2 = 270 \text{ mA}$

For CCM $\Rightarrow I_o > I_b \Rightarrow I_o > 270 \text{ mA}$

Observations:

For R load = 50 ohms

$I_o = 424 \text{ mA}$

$V_o = 21.4 \text{ V}$

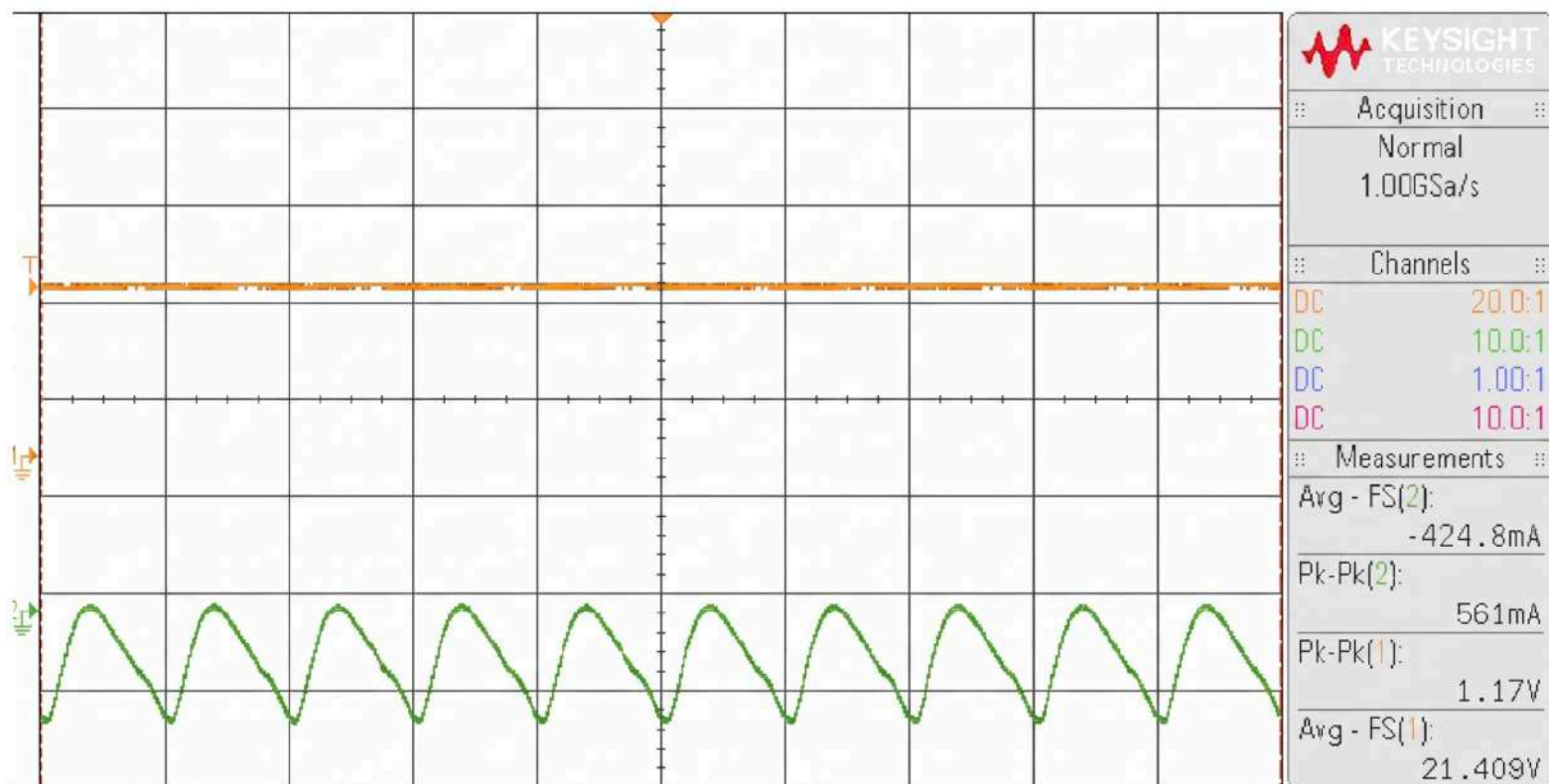


Fig: Orange Waveform : Output Voltage
Green Waveform : Inductor current